AdaptConfig

AI-Powered Integration Configuration Platform
for Enterprise Lending

Team Nucleolus · IIT Patna

S Akash (Lead) · Swayam Jain · Yash Kamdar

FinSpark Hackathon · April 2026

https://adaptconfig-frontend-production.up.railway.app

The Problem

- Credit Bureaus (CIBIL)
- Aadhaar eKYC
- GST Verification
- Payment Gateways
- Fraud Detection
- SMS / Email / AA

Pain Points

- **2–4 weeks** per integration
- Field mapping errors → data loss
- No pre-production testing
- 8 APIs = **16–32 weeks total**

Cost: 8 APIs × Rs 1–2L each = **Rs 8–16 Lakhs**

Our Solution

Upload API Spec $\xrightarrow{\text{AI}}$ **Auto-Config** $\xrightarrow{\text{Test}}$ **Deploy**

From **weeks** to **minutes**. Zero manual field mapping.

Metric	Result	
Endpoints parsed	4	from CIBIL v2 spec
Fields extracted	28	with types & required flags
Auth schemes	2	OAuth2 + API Key
Parse confidence	95%	
Fields mapped	28/28	at 100% confidence
Simulation	8/8 PASSED	all steps green
Total time	¡ 5 seconds	end to end

Architecture

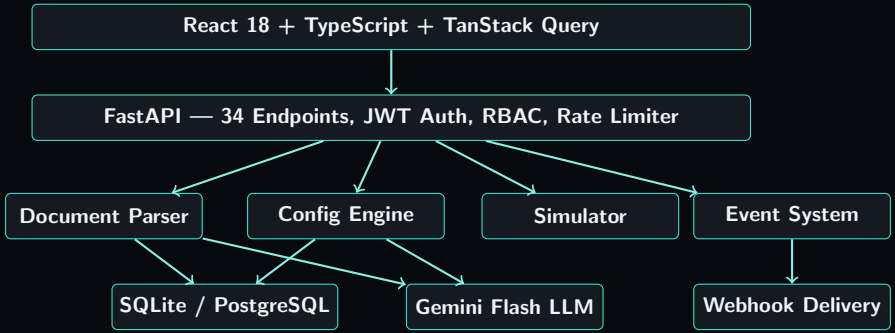
React Frontend $\xrightarrow{\text{JWT}}$ **FastAPI Backend (34 endpoints)** $\rightarrow$ **PostgreSQL**

Layer	Technologies
Frontend	React 18 · TypeScript · Tailwind CSS · Recharts
Backend	FastAPI · SQLAlchemy (async) · Pydantic v2
AI Engine	Google Gemini 3 Flash (REST API)
Services	Document Parser · Config Engine · Simulator
Database	PostgreSQL (prod) · SQLite (dev)
Deploy	Railway · Docker · nginx · Alembic migrations
Auth	JWT · PBKDF2 · RBAC · per-user tenant isolation

Backend serves 34 API endpoints across 10 route modules.

AI pipeline: Gemini 3 generates config $\rightarrow$ Rule engine validates with fuzzy matching.

Architecture Deep-Dive



Config Generation Pipeline

Hybrid AI + Rule Engine:

1. Gemini Flash generates config from document + adapter schema
2. **Rule engine** augments with:
   - 100+ Indian fintech synonyms
   - Fuzzy string matching (rapidfuzz)
   - Jaccard token overlap
3. Rule-based confidence **overrides** LLM
4. Unmapped fields **backfilled**

Graceful degradation:

Path	When
LLM + Rules	API key present
Rule fallback	LLM fails
Pure rules	AI disabled

Field matching tiers:

1. Exact synonym → 1.0
2. Fuzzy match → 0.6–0.99
3. Token Jaccard → 0.6+

Rollback & Version Control

Configuration Lifecycle:

$$\text{Draft} \rightarrow \text{Configured} \rightarrow \text{Validating} \rightarrow \text{Testing} \rightarrow \text{Active}$$

Version history:

- Every change creates a snapshot
- Full state serialization
- One-click rollback to any version
- Pre-rollback snapshot for safety

Diff engine:

- Recursive dict/list comparison
- Breaking change detection
- Side-by-side version comparison
- Export as JSON or YAML

Audit trail:

- Immutable log of every action
- Actor, timestamp, resource details
- Per-user data isolation

Database Schema (11 Tables)

Table	Purpose	Key Relations
`users`	Auth, RBAC (admin/editor/viewer)	tenant owner
`documents`	Uploaded specs (PDF/DOCX/YAML)	tenant-scoped
`adapters`	9 pre-built adapters	global catalog
`adapter_versions`	Multi-version per adapter	FK → adapters
`configurations`	Generated integration configs	FK → adapter, doc
`config_history`	Versioned snapshots for rollback	FK → configs
`simulations`	Test run results	FK → configs
`simulation_steps`	Per-step test details	FK → simulations
`audit_logs`	Immutable action log	tenant-scoped
`webhooks`	Registered callback URLs	tenant-scoped
`webhook_deliveries`	Delivery attempts & status	FK → webhooks

UUID primary keys · Row-level tenant isolation · Alembic migrations · CASCADE deletes

Authentication & User Isolation

Security:

- JWT access (30min) + refresh (7d) tokens
- PBKDF2-SHA256 password hashing with salt
- Per-user data isolation (unique tenant ID)
- Role-based access: admin / editor / viewer

Each user sees only their own:

Documents, Configs, Simulations, Audit

Adapters are shared (global catalog).

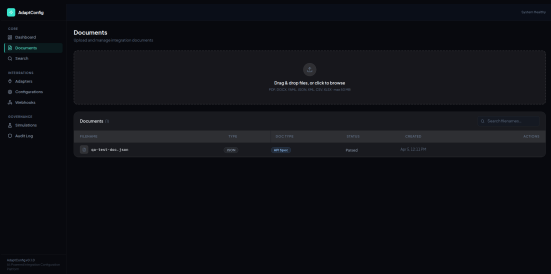
Additional protections:

- CORS restricted to known origins
- CSP, HSTS, X-Frame headers
- Rate limiting (100 req/min)
- PII masking (Aadhaar, PAN, phone)
- AES-256 Fernet encryption
- Path traversal protection
- Alembic database migrations
- Structured JSON logging

Dashboard

AdaptConfig
System Healthy
CORE
Dashboard
Documents
Search
INTEGRATIONS
Adapters
Configurations
Webhooks
GOVERNANCE
Simulations
Audit Log
S Akash
Sign out
Configuration Summary
1 total configuration
BY STATUS
Configured
1
AVG. CONFIDENCE
64%
across all mapped configurations
Weekly Activity
Documents & configurations processed
3
2.25
1.5
0.75
0
Mon
Tue
Wed
Thu
Fri
Sat
Sun
Documents
Configs
Adapter Status
Connected adapters
Active
8
Data Throughput
Records processed per hour
4 processed
2
1.5
1
0.5
0
00:00
04:00
08:00
12:00
16:00
20:00

Step 1: Upload & Parse



Formats: PDF, DOCX, YAML, JSON

Parser extracts:

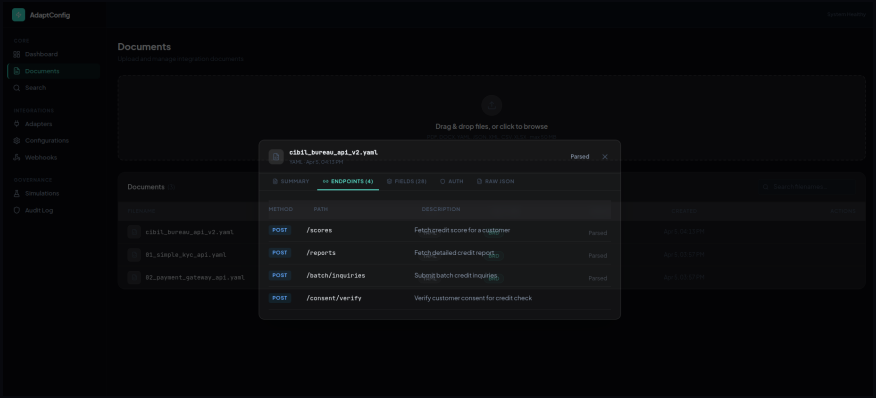
- API endpoints & methods
- Request/response fields
- Auth schemes
- SLA requirements

CIBIL v2 result:

4 endpoints · 28 fields · 2 auth

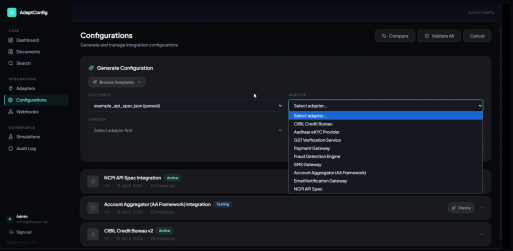
95% confidence

Parsed Document — 5-Tab Detail View



Summary · Endpoints · Fields (28) · Auth · Raw JSON

Step 2: AI Config Generation



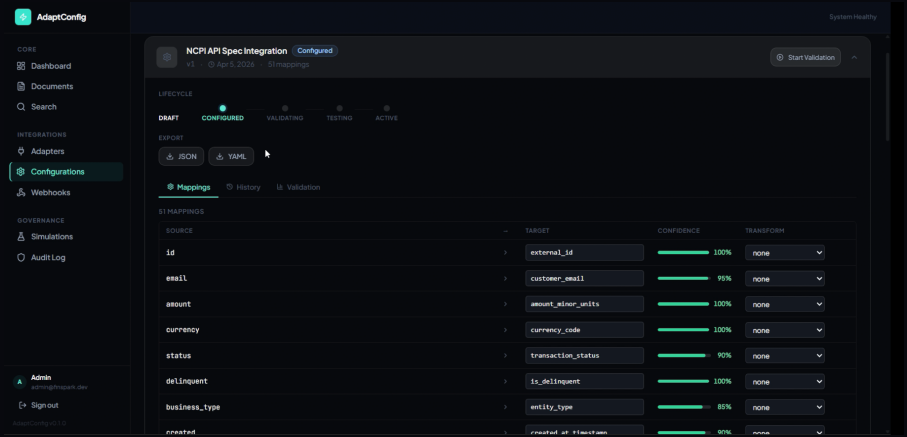
Two-stage pipeline:

1. Gemini 3 Flash generates config from document + adapter schema
2. **Rule engine** augments with fuzzy matching, Indian fintech synonyms, confidence scores

28/28 fields mapped at 100%

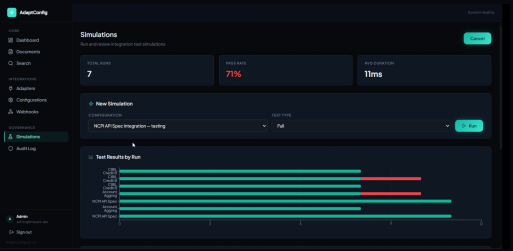
If no adapter matches, auto-detects and prompts to create one from the spec.

Editable Field Mappings



Lifecycle stepper · Editable targets · Transform dropdowns · Confidence bars · Export · History

Step 3: Simulation

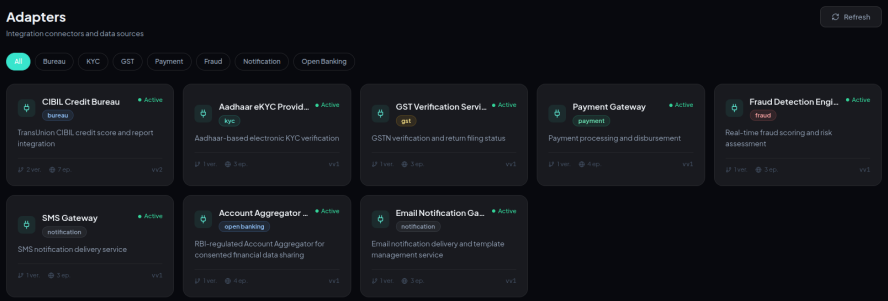


8-step pipeline:

- ✓ Config structure
- ✓ Field mapping coverage
- ✓ Endpoint: /scores
- ✓ Endpoint: /reports
- ✓ Endpoint: /batch/inquiries
- ✓ Endpoint: /consent/verify
- ✓ Auth validation
- ✓ Hooks validation

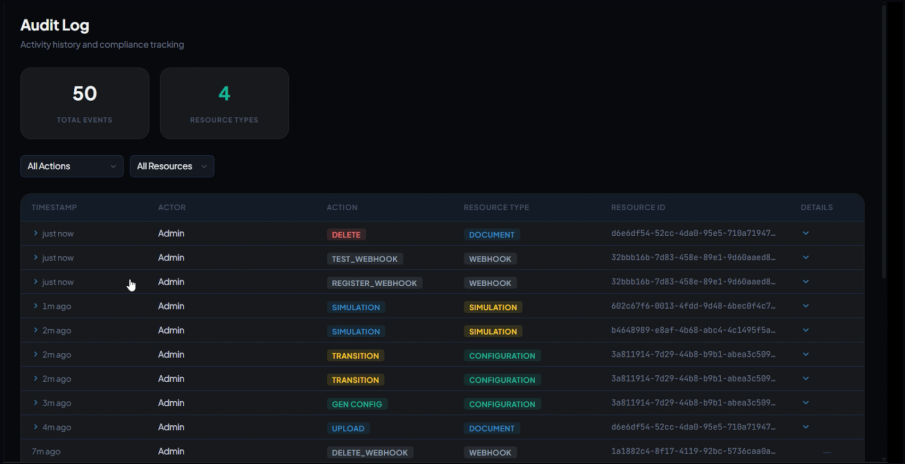
8/8 PASSED

Adapter Catalog (9 Pre-Built + Custom)



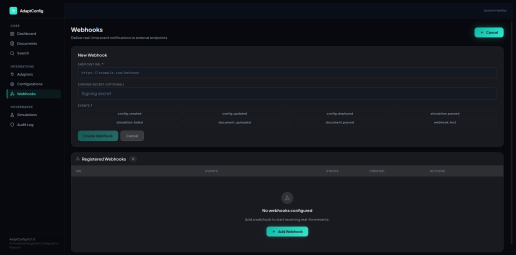
CIBIL · eKYC · GST · Payment · Fraud · SMS · Account Aggregator · Email · Custom

Audit Trail



Immutable log of every action · Filter by action type & resource · Pagination · Per-user isolation

Webhooks & Event System



Event-driven notifications:

- Register webhook URLs for real-time callbacks
- Subscribe to events: config.created, simulation.completed, document.uploaded
- Test delivery with one click
- HMAC signature verification
- Delivery status tracking

Use cases:

- CI/CD pipeline triggers
- Slack/Teams notifications

Results

	Manual	AdaptConfig	Improvement
Time per integration	2–4 weeks	$<$ 5 minutes	$\sim$500x faster
Field mapping errors	Common	AI-validated	Near zero
Pre-deploy testing	Rare	8-step pipeline	Always
Cost per platform	Rs 8–16L	Minimal	$>$90% savings

Metric	Value
API Endpoints	34
Adapters	9
Tests	899
Coverage	82%
Pages	8

Performance	Value
Parse confidence	95%
Mapping accuracy	100%
Simulation	8/8 pass
Config gen	$<$ 5s
Doc parse	$<$ 1s

Live Demo

https://adaptconfig-frontend-production.up.railway.app

1. **Register / Login**
   Create account or use demo credentials
2. **Upload API Spec**
   Drag & drop YAML, PDF, or DOCX
3. **View Parsed Fields**
   28 fields, 4 endpoints, 2 auth schemes
4. **Generate Config**
   Gemini 3 + Rule Engine pipeline
5. **Review Mappings**
   Edit targets, transforms, confidence bars
6. **Simulate & Test**
   8/8 steps pass with mock API responses
7. **Deploy**
   Configured → Validating → Testing → Active

Default admin: admin@finspark.dev / Admin1234!

LLM-Powered NLP Architecture

4 services enhanced with Gemini 2.5 Flash:

1. Document Parser – LLM extracts endpoints, fields, auth, SLA from *any* doc type (not just BRD/SOW)
2. Field Mapper – Semantic matching replaces fuzzy string + synonym dicts
3. Simulation Validator – LLM analyzes config quality with explanations
4. Search Service – NLU query understanding replaces keyword dicts

Design principles:

- Every LLM call has a **rule-based fallback**
- Gated by `FINSPARK_AI_ENABLED` flag
- Zero LLM cost when AI is disabled
- JSON structured output via `responseMimeType`
- Temperature 0.1 for deterministic results
- All calls augmented with regex-detected entities

LLM Cost Analysis – Gemini 2.5 Flash

LLM Call	Input (tok)	Output (tok)	Cost/call
Document Parsing	4,000	2,000	$0.008
Config Generation	3,000	2,000	$0.008
Field Mapping	1,000	1,000	$0.004
Simulation Validation	2,000	2,000	$0.008
Search Query	500	200	$0.001
Full workflow	**10,500**	**7,200**	**$0.028**

Monthly cost @ thinking ON:

100 users: $39/mo ($0.39/user)

1,000 users: $394/mo ($0.39/user)

With thinking OFF (-78%):

100 users: $8.50/mo ($0.09/user)

1,000 users: $85/mo ($0.09/user)

Pricing: Gemini 2.5 Flash – $0.30/M in, $3.50/M out (thinking ON) | $0.15/M in, $0.60/M out (thinking OFF)

Thank You

Questions?

Live Demo	https://adaptconfig-frontend-production.up.railway.app
API Docs	https://adaptconfig-api-production.up.railway.app/docs
Source Code	https://github.com/Akasxh/adaptconfig

Team Nucleolus

S Akash · Swayam Jain · Yash Kamdar

Indian Institute of Technology, Patna